

# Projections of self-affine sets onto lines

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This talk exhibits a recent work  
with Balázs Bárány and István Kolossváry

# Marstrand–Mattila projection theorem

For  $V \in \mathbb{RP}^{d-1}$ , let  $\text{proj}_V: \mathbb{R}^d \rightarrow V$  be the orthogonal projection.

**Theorem** (Marstrand (1954) and Mattila (1975)). *For every Borel set  $X \subseteq \mathbb{R}^d$  and  $\gamma_{d,1}$ -almost every  $V \in \mathbb{RP}^{d-1}$ ,*

$$\dim_{\text{H}}(\text{proj}_V(X)) = \min\{1, \dim_{\text{H}}(X)\}.$$

Mattila–Kaufman (1975) proved that the Hausdorff dimension  $\dim_{\text{H}}$  of the exceptional set can be as large as possible. Therefore, for sets without internal symmetries this is the whole story.

**Guiding question.** For **structured** sets  $X$ , can we describe the *exceptional set*

$$\mathcal{E}(X) = \{V \in \mathbb{RP}^{d-1} : \dim_{\text{H}}(\text{proj}_V(X)) < \min\{1, \dim_{\text{H}}(X)\}\}?$$

# Self-affine sets and irreducibility

An *affine iterated function system* on  $\mathbb{R}^d$  is a tuple  $\Phi = (\varphi_1, \dots, \varphi_N)$  with maps  $\varphi_i(x) = A_i x + t_i$ , where  $A_i \in \text{GL}(d, \mathbb{R})$  and  $\|A_i\| < 1$ . There is a unique non-empty compact  $X \subset \mathbb{R}^d$  such that

$$X = \bigcup_{i=1}^N \varphi_i(X).$$

We write  $A = (A_1, \dots, A_N)$  for the tuple of linear parts and  $A_{\mathbf{i}} = A_{i_1} \cdots A_{i_n}$  for the product associated to a word  $\mathbf{i} = i_1 \cdots i_n$ .

**Definition.** We say that  $A \in \text{GL}(d, \mathbb{R})^N$  is

- ▶ *reducible* if there is a proper non-trivial linear subspace  $V \subset \mathbb{R}^d$  with  $A_i V = V$  for all  $i$ ,
- ▶ *strongly irreducible* if no finite union of proper non-trivial linear subspaces of  $\mathbb{R}^d$  is invariant under every  $A_i$ ,
- ▶ *proximal* if some  $A_{\mathbf{i}}$  has a simple dominant eigenvalue.

# Exceptional sets in $\mathbb{R}^2$ : state of the art

**Falconer–Kempton (2017).** For many dominated self-affine sets  $X \subset \mathbb{R}^2$ ,  $\#\mathcal{E}(X) \leq 1$ .

**Bárány–Hochman–Rapaport (2019).** For proximal, strongly irreducible self-affine sets  $X \subset \mathbb{R}^2$  satisfying the strong open set condition,  $\mathcal{E}(X) = \emptyset$ .

**Hochman–Rapaport (2021).** The strong open set condition in the result above can be relaxed to exponential separation.

**Bárány–Käenmäki–Pyörälä–Wu (2023+) and Pyörälä (2025).** For self-affine measures on planar carpets, at most two exceptional directions; Pyörälä removes the separation assumption.

# Exceptional sets in $\mathbb{R}^d$ : state of the art

**Feng–Xie (2025) and Morris–Sert (2025).** Under **irreducibility**,  $\mathcal{E}(X) = \emptyset$  for **Lebesgue-almost every** choice of translations  $t_1, \dots, t_N$ .

Here  $\dim_{\text{aff}}$  is *affinity dimension*—an symbolic upper bound for  $\dim_H$ .

**Feng–Xie (2025) and Morris–Sert (2025).** For **reducible**  $X \subset \mathbb{R}^d$  such that some proper  $A^\top$ -invariant subspace  $W$  has  $\dim_{\text{aff}}(A^\top|_W) < \min\{1, \dim_{\text{aff}}(A)\}$ , there is  $V \in \mathbb{RP}^{d-1}$  with  $V \subseteq W$  and

$$\overline{\dim}_M(\text{proj}_V(X)) < \min\{1, \dim_H(X)\}$$

for **Lebesgue-almost every** choice of translations  $t_1, \dots, t_N$ .

**Question.** Which self-affine sets  $X \subset \mathbb{R}^d$  satisfy  $\mathcal{E}(X) = \emptyset$  in every dimension  $d$ , for every choice of translations  $t_1, \dots, t_N$ , and without separation assumptions?

# Corollaries of our main theorem

Although not stated explicitly in Hochman–Rapaport (2021), the following result follows from their work via the arguments of Bárány–Hochman–Rapaport (2019).

**Theorem** (Hochman–Rapaport (2021)). *Let  $X \subset \mathbb{R}^2$  be a proximal, strongly irreducible self-affine set satisfying the *exponential separation condition*. Then for every  $V \in \mathbb{RP}^1$ ,*

$$\dim_{\mathrm{H}}(\mathrm{proj}_V(X)) = \min\{1, \dim_{\mathrm{H}}(X)\}.$$

A corollary of our main theorem removes the exponential separation hypothesis:

**Corollary 1.** *Let  $X \subset \mathbb{R}^2$  be a proximal, strongly irreducible self-affine set. Then for every  $V \in \mathbb{RP}^1$ ,*

$$\dim_{\mathrm{H}}(\mathrm{proj}_V(X)) = \min\{1, \dim_{\mathrm{H}}(X)\}.$$

# Corollaries of our main theorem

Here  $\dim_M(X)$  is the *Minkowski dimension*—the growth rate of the number of  $\delta$ -balls needed to cover  $X$  as  $\delta \rightarrow 0$ , always  $\geq \dim_H(X)$  which allows covers by sets of varying radii.

**Theorem** (Hochman–Rapaport (2021)). *Let  $X \subset \mathbb{R}^2$  be a proximal, strongly irreducible self-affine set satisfying the exponential separation condition. Then*

$$\dim_H(X) = \dim_M(X) = \dim_{\text{aff}}(X).$$

A corollary of our main theorem removes the exponential separation hypothesis when the upper Minkowski dimension is at most one:

**Corollary 2.** *Let  $X \subset \mathbb{R}^2$  be a proximal, strongly irreducible self-affine set with  $\overline{\dim}_M(X) \leq 1$ . Then*

$$\dim_H(X) = \dim_M(X).$$

# Main Theorem

**Main Theorem.** Let  $X \subset \mathbb{R}^d$  be a *Zariski dense* self-affine set. Then, for every  $V \in \mathbb{RP}^{d-1}$ ,

$$\dim_H(\text{proj}_V(X)) = \min\{1, \dim_H(X)\} = \min\{1, \overline{\dim}_M(X)\}.$$

- ▶ **Corollary 1** follows once we show that proximality and strong irreducibility imply Zariski density for  $A \in \text{GL}(2, \mathbb{R})^N$ .
- ▶ **Corollary 2** then follows trivially from

$$\begin{aligned} \underline{\dim}_M(X) &\geq \dim_H(X) \geq \dim_H(\text{proj}_V(X)) \\ &= \min\{1, \overline{\dim}_M(X)\} = \overline{\dim}_M(X). \end{aligned}$$

It also provides a *partial positive answer* to the folklore question:

**Question (Folklore).** Does the Minkowski dimension of every self-affine set exist?



# Zariski density

Let  $\mathcal{S}(A) = \langle A_1, \dots, A_N \rangle$  be the subsemigroup generated by  $A$ .

**Definition.** A tuple  $A = (A_1, \dots, A_N) \in \mathrm{GL}(d, \mathbb{R})^N$  is **Zariski dense** if the Zariski closure of  $\mathcal{S}(A)$  in  $\mathrm{GL}(d, \mathbb{R})$  is all of  $\mathrm{GL}(d, \mathbb{R})$ .

Equivalently, every polynomial in the matrix entries and  $1/\det$  that vanishes on  $\mathcal{S}(A)$  vanishes on all of  $\mathrm{GL}(d, \mathbb{R})$ : the matrices satisfy no nontrivial polynomial relation, and the semigroup is algebraically as large as possible.

**Two consequences.** If  $A$  is Zariski dense, then

- 1**  $A$  is strongly  $k$ -irreducible for every  $k \in \{1, \dots, d-1\}$ ,
- 2**  $A$  is strongly pinching.

Both lift the strong irreducibility and proximality of  $A$  to the exterior powers  $\wedge^k \mathbb{R}^d$ .

# Strong $k$ -irreducibility and strong pinching

For  $k \in \{1, \dots, d-1\}$ , each  $A \in \mathrm{GL}(d, \mathbb{R})$  induces a linear map  $A^{\wedge k}$  on the exterior power  $\wedge^k \mathbb{R}^d$ , whose eigenvalues are the  $k$ -fold products of distinct eigenvalues of  $A$ . Intuitively,  $A^{\wedge k}$  records how  $A$  distorts  $k$ -dimensional parallelepipeds.

**Definition.** We say that  $A \in \mathrm{GL}(d, \mathbb{R})^N$  is

- ▶ *strongly  $k$ -irreducible* if no finite union of proper non-trivial subspaces of  $\wedge^k \mathbb{R}^d$  is invariant under every  $A_i^{\wedge k}$ ,
- ▶ *strongly pinching* if some single  $A \in \mathcal{S}(A)$  has every  $A^{\wedge k}$  diagonalizable with eigenvalues of pairwise distinct moduli.

Strong 1-irreducibility recovers strong irreducibility. **Strong pinching sharpens proximality**: distinct moduli force a simple dominant eigenvalue, so a single  $A^{\wedge k}$  witnesses proximality on every exterior power  $\wedge^k \mathbb{R}^d$  at once.

# When does Zariski density hold?

The following proposition verifies Corollaries 1 and 2.

**Proposition.** *For contractive  $A \in \mathrm{GL}(2, \mathbb{R})^N$ , the tuple  $A$  is Zariski dense if and only if  $A$  is 1-proximal and strongly 1-irreducible.*

- ▶ In  $d = 2$ , the proper algebraic subgroups of  $\mathrm{PGL}(2, \mathbb{R})$  are few—diagonal, upper triangular,  $\mathrm{SO}(2)$ , and finite groups—and proximality together with strong irreducibility rules each out, while contractivity supplies the missing determinant information. Without contractivity the equivalence fails in every  $d$ :  $\mathrm{SL}(d, \mathbb{R})$  is a proper algebraic subgroup of  $\mathrm{GL}(d, \mathbb{R})$ .
- ▶ In  $d \geq 3$  the equivalence breaks: there exist contractive tuples that are strongly pinching and strongly  $k$ -irreducible for every  $k \in \{1, \dots, d-1\}$ , yet preserve an indefinite quadratic form and are confined to its orthogonal group up to scaling. Hence Zariski density is **strictly stronger** than strong pinching plus strong  $k$ -irreducibility in dimension  $d \geq 3$ .

# When does Zariski density hold?

**Proposition.** *Zariski dense tuples form an **open and dense** subset of  $\mathrm{GL}(d, \mathbb{R})^N$ .*

A tuple fails to be Zariski dense only when its semigroup is confined to a proper algebraic subgroup of  $\mathrm{GL}(d, \mathbb{R})$ , such as one preserving a quadratic form or a subspace, or with determinants in  $\{1, -1\}$ .

**Sharpness.** Feng–Xie (2025) and Morris–Sert (2025) construct reducible self-affine sets with non-trivial exceptional set at Lebesgue-typical translations, so the conclusion of Main Theorem genuinely requires Zariski density. Whether it is the **minimal** algebraic hypothesis—in particular, whether e.g. irreducible quadratic-form-preserving tuples in  $d \geq 3$  admit an all-directions theorem—remains open.

Let us next sketch the proof of the main theorem.

# Step 1: the measure theorem (the engine)

- ▶ The proof of the main theorem routes through self-affine measures and reduces to subsystems. The first step is the measure analogue.
- ▶ A fully supported Bernoulli measure  $\nu$  on  $\mathcal{I}^{\mathbb{N}} = \{1, \dots, N\}^{\mathbb{N}}$  pushes forward through the natural coding  $\pi: \mathcal{I}^{\mathbb{N}} \rightarrow X$  to a *self-affine measure*  $\mu = \pi_*\nu$ . Its *Lyapunov dimension*  $\dim_L(\nu)$  is determined by the Kolmogorov–Sinai entropy of  $\nu$  and the Lyapunov exponents of  $A$ .

**Theorem 3.** *Let  $\nu$  be a fully supported Bernoulli measure and  $\mu$  the associated self-affine measure. Suppose  $\Phi$  satisfies the *exponential separation condition* and  $A$  is  *$k$ -proximal* and *strongly  $k$ -irreducible* for every  $k \in \{1, \dots, d-1\}$ . Then, for every  $V \in \mathbb{RP}^{d-1}$ ,*

$$\dim((\text{proj}_V)_*\mu) = \min\{1, \dim_L(\nu)\}.$$

Let us next define the symbolic dimensions  $\dim_{\text{aff}}(A)$  and  $\dim_L(\nu)$ .

# Symbolic dimensions

To make  $\dim_{\text{aff}}(A)$  and  $\dim_L(\nu)$  precise, we use Falconer's *singular value function*. For  $A \in \text{GL}(d, \mathbb{R})$ , let  $\alpha_1(A) \geq \cdots \geq \alpha_d(A) > 0$  be the singular values—the semi-axes of the ellipsoid  $A(B(0, 1))$ —and for  $0 \leq s \leq d$  with  $k = \lfloor s \rfloor$ , set

$$\varphi^s(A) = \alpha_1(A) \cdots \alpha_k(A) \alpha_{k+1}(A)^{s-k}.$$

The *affinity dimension*  $\dim_{\text{aff}}(A)$  is the unique  $s \in [0, d]$  with

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \sum_{\mathbf{i} \in \mathcal{I}^n} \varphi^s(A_{\mathbf{i}}) = 0,$$

or  $d$  if no such  $s$  exists, and depends only on  $\Phi$ . For a Bernoulli measure  $\nu$  on  $\mathcal{I}^{\mathbb{N}}$ , the *Lyapunov dimension*  $\dim_L(\nu)$  is the unique  $s$  with  $h(\nu) + \Lambda(A, \nu, s) = 0$ , where  $h(\nu)$  is the entropy of  $\nu$  and

$$\Lambda(A, \nu, s) = \lim_{n \rightarrow \infty} \frac{1}{n} \int \log \varphi^s(A_{\mathbf{i}|_n}) d\nu(\mathbf{i}).$$

Both bound the geometric dimension from above: Falconer (1988) proved  $\overline{\dim}_M(X) \leq \dim_{\text{aff}}(A)$ , and always  $\dim_L(\nu) \leq \dim_{\text{aff}}(A)$ .

## Step 2: subsystems

Theorem 3 needs **exponential separation** and  **$k$ -proximality**, **strong  $k$ -irreducibility** for every  $k$ . The main theorem assumes neither. We extract a subsystem  $\Phi' \subseteq \Phi$  that has both.

**Proposition 4.** *For every  $\varepsilon > 0$  there is a finite collection of words  $\mathcal{J} \subset \mathcal{I}^*$  such that  $\Phi' = (\varphi_i)_{i \in \mathcal{J}}$*

- ▶ *satisfies the **strong separation condition** (hence exponential separation),*
- ▶ *is **dominated**:  $\|A_{ij}\| \geq \kappa \|A_i\| \|A_j\|$  on  $\mathcal{J}^*$  for some  $\kappa \in (0, 1]$ ,*
- ▶ *has  **$k$ -proximal** and **strongly  $k$ -irreducible** for every  $k$ ,*
- ▶ *has  $\dim_{\text{aff}}(A') \geq \min\{1, \overline{\dim}_M(X)\} - \varepsilon$ .*

The proof of Proposition 4 proceeds in **two steps**: first we construct a dominated subsystem, then a strongly separated packing inside it.

## Step 3: transfer to sets

Fix  $\varepsilon > 0$  small, take  $\Phi'$  and  $\mathcal{J}$  from [Proposition 4](#), and set

$$s = \min\{1, \dim_{\text{aff}}(A')\} - \varepsilon \in (0, 1).$$

Choose  $n \geq 1$  with  $\kappa^s \sum_{i \in \mathcal{J}^n} \|A_i\|^s > 1$ , set  $\mathcal{K} = \mathcal{J}^n$  and  $\Phi'' = (\varphi_i)_{i \in \mathcal{K}}$ , and let  $\nu$  be the Bernoulli measure on  $\mathcal{K}^{\mathbb{N}}$  with weights  $p_i \approx \|A_i\|^s$ . The super-multiplicativity from domination then yields

$$\dim_L(\nu) \geq s.$$

Since  $\mathcal{K} = \mathcal{J}^n$ , the subsystem  $\Phi''$  inherits the strong separation condition from  $\Phi'$  and is  $k$ -proximal and strongly  $k$ -irreducible for every  $k$ . By [Theorem 3](#),

$$\begin{aligned} \dim_H(\text{proj}_V(X)) &\geq \dim((\text{proj}_V)_* \mu) = \min\{1, \dim_L(\nu)\} \\ &\geq \min\{1, \dim_{\text{aff}}(A')\} - \varepsilon \geq \min\{1, \overline{\dim}_M(X)\} - 2\varepsilon. \end{aligned}$$

Letting  $\varepsilon \rightarrow 0$  finishes the proof of the Main Theorem assuming [Steps 1 and 2](#). □



# Step 1: proof outline

We now sketch the proof of Theorem 3, the measure theorem of [Step 1](#). The argument proceeds in two steps.

**Step 1.1** (Furstenberg-typical directions). We establish the conclusion at directions  $V$  typical for a natural [stationary measure](#)  $\mu_F$  on  $\mathbb{RP}^{d-1}$  associated with  $A^\top$  and  $\nu$ , by combining the entropy-dimension formula of Rapaport (2024) with Feng's (2023) exact-dimensionality theorem.

**Step 1.2** (upgrade to every direction). We pass from  $\mu_F$ -typical  $V$  to every prescribed  $V \in \mathbb{RP}^{d-1}$ . The catch is that  $\mu_F$  can be singular with respect to the spherical measure, with Cantor-like support missing the directions we want. A suspension semiflow then transports the typical-direction estimate to [any](#) such  $V$ .

# Exponential separation

The measure theorem requires a quantitative no-overlap condition.

**Definition.** We say that  $\Phi = (\varphi_1, \dots, \varphi_N)$  satisfies the *exponential separation condition* if there exist  $c > 0$  and infinitely many  $n \geq 1$  such that for all distinct  $i \neq j \in \mathcal{I}^n$ ,

$$\|\varphi_i - \varphi_j\|_\infty \geq c^n.$$

Exponential separation is the standard hypothesis in the literature for dimension theorems on self-affine measures. Even for self-similar sets on the real line, dimension theory without it is wide open—this is the celebrated *dimension drop conjecture*.

**Intuition.** Exponential separation is mild: it holds for any IFS with algebraic entries and no exact overlaps (Gamburd–Jakobson–Sarnak (1999) for  $d = 2$ , generalized here to all  $d$ ), and in  $\mathbb{R}$  for Lebesgue-almost every choice of translations (Hochman (2014)).

# The Furstenberg measure

When  $A$  is 1-proximal and strongly 1-irreducible, there is a unique probability measure  $\mu_F$  on  $\mathbb{RP}^{d-1}$ , the *Furstenberg measure*, satisfying the stationarity equation

$$\mu_F = \sum_{i=1}^N p_i (A_i^\top)_* \mu_F, \quad p_i = \nu([i]).$$

**Intuition.** Consider the random walk on  $\mathbb{RP}^{d-1}$  that, at each step, draws  $i \in \mathcal{I}$  from  $(p_i)_i$  and applies  $A_i^\top$  to the current direction. The Furstenberg measure  $\mu_F$  is the unique stationary distribution of this walk, and iterates of any starting direction converge in law to  $\mu_F$  (Furstenberg (1963)).

**Why  $A^\top$  and not  $A$ ?** Pulling  $V$  back through  $\text{proj}_V \circ \varphi_i$  gives  $A_i^\top V$ , since  $\langle A_i x, V \rangle = \langle x, A_i^\top V \rangle$ . Hence the natural action on projection directions is through the transposed semigroup.

# Step 1.1: Furstenberg-typical projection

**Theorem 5** (Rapaport (2024) and Feng (2023)). *Let  $\nu$  be a fully supported Bernoulli measure and  $\mu$  the associated self-affine measure. Suppose  $\Phi$  satisfies the *exponential separation condition* and  $A$  is *k-proximal* and *strongly k-irreducible* for every  $k \in \{1, \dots, d-1\}$ . Then,  $\mu_F$ -almost every  $V \in \mathbb{RP}^{d-1}$ ,*

$$\dim((\text{proj}_V)_*\mu) = \min\{1, \dim_L(\nu)\}.$$

The proof of Theorem 5 combines two ingredients.

- ▶ Rapaport (2024) supplies the *entropy-dimension* identity for  $(\text{proj}_V)_*\mu$  at every  $V \in \mathbb{RP}^{d-1}$ .
- ▶ Feng (2023) upgrades the entropy dimension to *exact dimension* (Hausdorff = packing = entropy) on a set of full  $\mu_F$ -measure.

Combining the two yields the conclusion at a  $\mu_F$ -typical  $V$ —this is **Step 1.1**.

## Step 1.2: from $\mu_F$ -typical to every direction

Why is **Theorem 5** not enough? We need an estimate at a **prescribed**  $V$ , not a typical one. The Furstenberg measure  $\mu_F$  can be singular, even supported on a Cantor-like subset of  $\mathbb{RP}^{d-1}$  that misses most directions.

**Step 1.2.1** (suspension semiflow). Build a **suspension semiflow** on  $\mathcal{I}^{\mathbb{N}} \times \mathbb{RP}^{d-1}$  whose direction coordinate **equidistributes** to  $\mu_F$  from any starting  $V$ .

**Step 1.2.2** (local entropy averages). Verify the **local-entropy-averages hypothesis** of Hochman–Shmerkin (2012) at  $V$ , transporting Rapaport’s typical-direction estimate.

**Step 1.2.3** (matching upper bound). Combine with Rossi’s (2014) **upper bound** to upgrade the lower bound to exact dimension.

## Step 1.2.1: suspension semiflow

Define a base map and a roof function on  $\mathcal{I}^{\mathbb{N}} \times \mathbb{RP}^{d-1}$  by

$$F(\mathbf{i}, V) = (\sigma \mathbf{i}, A_{i_1}^{\top} V), \quad r(\mathbf{i}, V) = -\log_2 \|A_{i_1}^{\top} V\|.$$

Identifying  $(\mathbf{i}, V, r(\mathbf{i}, V)) \sim (F(\mathbf{i}, V), 0)$ , the suspension space is

$$\mathcal{X} = \{(\mathbf{i}, V, t) \in \mathcal{I}^{\mathbb{N}} \times \mathbb{RP}^{d-1} \times \mathbb{R} : 0 \leq t < r(\mathbf{i}, V)\},$$

and the semiflow  $T_t$  increases the height  $t$  and folds at the roof.

**Equidistribution along orbits.** Since  $\mu_F$  is  $F$ -invariant, the normalized restriction

$$m = \frac{(\nu \times \mu_F \times \mathcal{L}^1)|_{\mathcal{X}}}{(\nu \times \mu_F \times \mathcal{L}^1)(\mathcal{X})}$$

is a  $T_t$ -invariant Borel probability on  $\mathcal{X}$ . Along  $T_t$ -orbits, the direction coordinate converges in law to  $\mu_F$  from **any** starting  $V \in \mathbb{RP}^{d-1}$ .

## Step 1.2.2: local entropy averages

**Theorem** (Hochman–Shmerkin (2012)). *Let  $\nu$  be a Borel probability on  $\mathcal{I}^{\mathbb{N}}$ ,  $f: \mathcal{I}^{\mathbb{N}} \rightarrow [-1, 1]$  continuous, and  $\{\mathcal{Q}_n^q\}$  a refining family of partitions of  $\mathcal{I}^{\mathbb{N}}$  with  $\text{diam}(f([i])) \approx 2^{-qn}$  on  $\mathcal{Q}_n^q$ . If, for  $\nu$ -almost every  $i$ ,*

$$\limsup_{q \rightarrow \infty} \liminf_{n \rightarrow \infty} \frac{1}{qn \log 2} \sum_{i=1}^n H(f_* \nu|_{[\mathcal{Q}_i^q(i)]}, \mathcal{D}_{q(i+1)}) \geq \alpha,$$

*then  $\underline{\dim}_H(f_* \nu) \geq \alpha$ .*

Apply this with  $f = \text{proj}_V \circ \pi$  and cutsets matched to  $\|A_i^\top|V\| \approx 2^{-qi}$ . The flow's  $T_q$ -steps match these cutset depths up to a bounded error, so [Rapaport's entropy bound](#) from [Step 1.1](#) verifies the local-entropy-averages hypothesis at the evolving direction; Furstenberg comparability then transfers it to the prescribed  $V$ , giving for every  $V \in \mathbb{RP}^{d-1}$ ,

$$\underline{\dim}_H((\text{proj}_V)_* \mu) \geq \min\{1, \dim_L(\nu)\}.$$

## Step 1.2.3: matching upper bound

**Proposition** (Rossi (2014)). *For any ergodic shift-invariant Borel probability  $\nu$  on  $\mathcal{I}^{\mathbb{N}}$  and the associated self-affine measure  $\mu = \pi_*\nu$ ,*

$$\overline{\dim}_p(\mu) \leq \min\{d, \dim_L(\nu)\}.$$

The projection  $\text{proj}_V: \mathbb{R}^d \rightarrow V$  is Lipschitz and lands on a line, so

$$\overline{\dim}_p((\text{proj}_V)_*\mu) \leq \min\{1, \overline{\dim}_p(\mu)\} \leq \min\{1, \dim_L(\nu)\}.$$

Together with the [Step 1.2.2](#) lower bound, for every  $V \in \mathbb{RP}^{d-1}$  we obtain

$$\dim((\text{proj}_V)_*\mu) = \min\{1, \dim_L(\nu)\},$$

upgrading [Theorem 5](#) from  $\mu_F$ -typical to all directions and completing [Step 1](#).



## Step 2: from measures to sets

Step 1 produced the **measure theorem**: under **exponential separation** and  **$k$ -proximity** plus **strong  $k$ -irreducibility** for every  $k$ ,

$$\dim((\text{proj}_V)_*\mu) = \min\{1, \dim_L(\nu)\}$$

at every  $V \in \mathbb{RP}^{d-1}$ . The Main Theorem asks the same conclusion for  $\dim_H(\text{proj}_V(X))$  from **Zariski density alone**—no separation, no algebraic hypotheses on  $A$  beyond Zariski density itself.

**Strategy.** Approximate  $X$  from inside by a subsystem  $X' \subseteq X$  such that  $\Phi'$  satisfies the **strong separation condition**,  $A'$  is  **$k$ -proximal** and **strongly  $k$ -irreducible** for every  $k$ , and  $\dim_{\text{aff}}(A') \geq \min\{1, \overline{\dim}_M(X)\} - \varepsilon$ . Apply **Step 1** to a self-affine measure on  $X'$  with  $\dim_L(\nu)$  close to  $\dim_{\text{aff}}(A')$ .

The subsystem is built in two stages—**Step 2.1** finds a dominated subsystem and **Step 2.2** makes it strongly separated.

## Step 2.1: dominated subsystem

**Proposition.** Let  $A \in GL(d, \mathbb{R})^N$  be 1-proximal and strongly 1-irreducible, and fix  $i \in \mathcal{I}^*$  with  $A_i$  proximal. There exist  $K \geq 1$  and maps  $\bar{h}, \bar{k}: \mathcal{I}^* \rightarrow \bigcup_{m=0}^K \mathcal{I}^m$  such that, for every finite  $\mathcal{D} \subset \mathcal{I}^*$ , the sandwiched tuple

$$A' = (A_{\bar{h}(j)} A_j A_{\bar{k}(j)})_{j \in \mathcal{D}}$$

is dominated.

This extends Bárány–Jordan–K–Rams (2021) from  $d = 2$  to all  $d$ .

**Strategy.** Take the cone  $\mathcal{C} = B(V, \varepsilon)$  around the leading eigenline  $V$  of  $A_i$ , with  $\mathcal{C}$  disjoint from the repelling hyperplane  $W$ . For each  $j$ , the suffix  $\bar{k}(j)$  steers the image of  $A_j A_{\bar{k}(j)} \mathcal{C}$  into one of finitely many cover sets  $Q_i$  of  $\mathbb{RP}^{d-1}$ ; the prefix  $\bar{h}(j)$  then uses an irreducibility word to push  $Q_i$  off  $W$  and powers of  $A_i$  to contract back into  $\mathcal{C}$ . Hence  $\mathcal{C}$  is a strongly invariant multicone for  $A'$ , and domination follows from Bochi–Gourmelon (2009).

## Step 2.2: strongly separated subsystem

**Proposition.** Let  $A \in \mathrm{GL}(d, \mathbb{R})^N$  be *Zariski dense*. For every  $\varepsilon > 0$  there is a finite  $\mathcal{J} \subset \mathcal{I}^*$  such that

- ▶  $\Phi' = (\varphi_j)_{j \in \mathcal{J}}$  satisfies the *strong separation condition*;
- ▶  $A' = (A_j)_{j \in \mathcal{J}}$  is dominated, and *k-proximal* and *strongly k-irreducible* for every  $k \in \{1, \dots, d-1\}$ ;
- ▶  $\min\{1, \overline{\dim}_M(X)\} - \varepsilon \leq \dim_{\mathrm{aff}}(A') < d$ .

The proof combines three ingredients:

- ▶ *Packing* yields  $\gtrsim r^{-\overline{\dim}_M(X) + \varepsilon}$  cylinders of comparable size at scale  $r$  with pairwise disjoint images.
- ▶ *Step 2.1* sandwiches each cylinder so that  $A'$  is dominated.
- ▶ *Zariski density* yields a *strongly pinching* word, which gives  $k$ -proximality, and *subspace avoidance* in every exterior power, which gives strong  $k$ -irreducibility.

# Finishing the proof

Steps 2.1 and 2.2 together prove Proposition 4, completing Step 2. As sketched earlier in Step 3, plugging strongly separated dominated subsystems of Proposition 4 into the measure projection theorem, Theorem 3, then yields

$$\dim_H(\text{proj}_V(X)) \geq \min\{1, \overline{\dim}_M(X)\} - 2\varepsilon$$

for every  $V \in \mathbb{RP}^{d-1}$ . Letting  $\varepsilon \rightarrow 0$  proves the Main Theorem:

**Main Theorem.** *Let  $X \subset \mathbb{R}^d$  be a Zariski dense self-affine set. Then, for every  $V \in \mathbb{RP}^{d-1}$ ,*

$$\dim_H(\text{proj}_V(X)) = \min\{1, \dim_H(X)\} = \min\{1, \overline{\dim}_M(X)\}.$$